Sample Project Idea #2

Maximizing detectability of liver lesions in low-dose CT with deep learning denoising

Brief project description:

Patient images in the data library were retrospectively obtained from clinically-indicated examinations. All data that were shared were fully anonymized. Cases included in the data library were either negative for findings in the liver or had focal liver lesions. Both benign and metastatic liver lesions were included. Potential cases were reviewed by a board-certified abdominal radiologist and excluded from the data library if there were more than 5 lesions or the lesion was greater than 3 cm. Reference data were gathered for each patient to provide a definitive diagnosis and establish the accuracy of host center's diagnosis.

Poisson noise was inserted into the projection data for each case in the library to reach a noise level that corresponded to 25% of the full dose (i.e. "quarter-dose" data were simulated). The projection data were from right before image reconstruction, i.e. after all preprocessing and taking the logarithm. For reconstruction algorithms requiring statistical information, a noise map, expressed as the noise-equivalent incident number of quanta for each data frame, was provided.

Full description of the project:
https://aapm.app.box.com/s/eaw4jddb53keg1bptavvvd1sf4x3pe9h

***Dataset*:**

The dataset involved in this project can be downloaded from Low Dose CT Grand Challenge

https://aapm.app.box.com/s/eaw4jddb53keg1bptavvvd1sf4x3pe9h

Note that the students will only use the Image Data with 1mm B30.

Objective:

The objective is to reduce image noise using image-domain deep learning techniques in order to maximize detectability of liver lesions in low-dose patient CT datasets.